ControlNet

Foreword

- This course introduces the ControlNet model and how to use it to control composition.

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Objectives

- Upon completion of this course, you will understand:
  - What is ControlNet
  - Common ControlNet models

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Contents

1. **Introduction to ControlNet**
2. Contour ControlNet Models
3. Depth ControlNet Models
4. Object ControlNet Models

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Introduction to ControlNet

- ControlNet is a neural network that controls pre-trained image diffusion models, such as Stable Diffusion. It allows users to input an adjustment image and use it to control image generation.
- At the International Conference on Computer Vision (ICCV) 2023, *Adding Conditional Control to Text-to-Image Diffusion Models* was awarded the best paper (Marr Prize).

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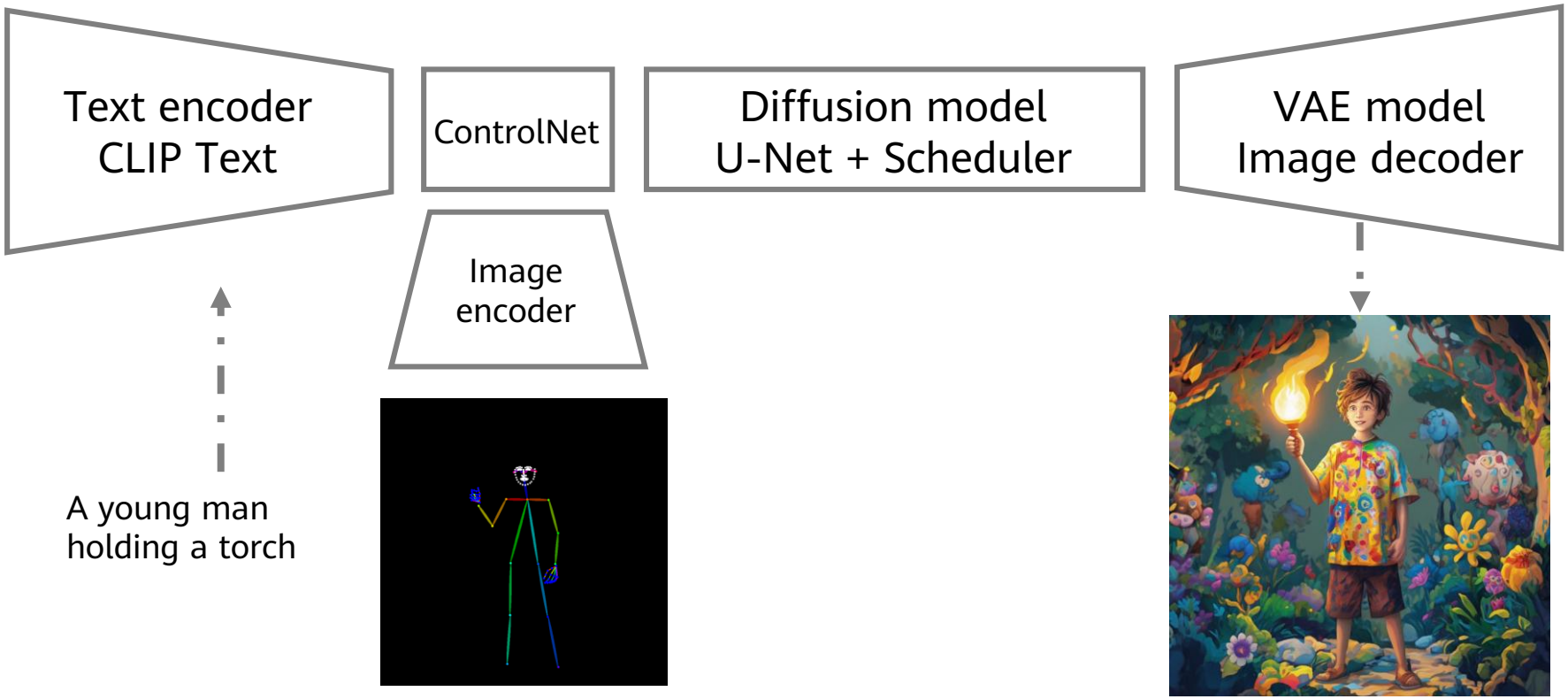
How ControlNet Works

- ControlNet enhances the functionality of image generation models, especially diffusion models such as Stable Diffusion.
- ControlNet guides diffusion models in converting text to images. While traditional image generation models can produce stunning visual effects, they usually lack guidance and cannot generate images that meet user requirements.



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ControlNet Technical Architecture

- ControlNet locks the original stable diffusion model, creates a trainable copy of the original model, and creates a zero convolution layer that uses the adjustment vector c as input.

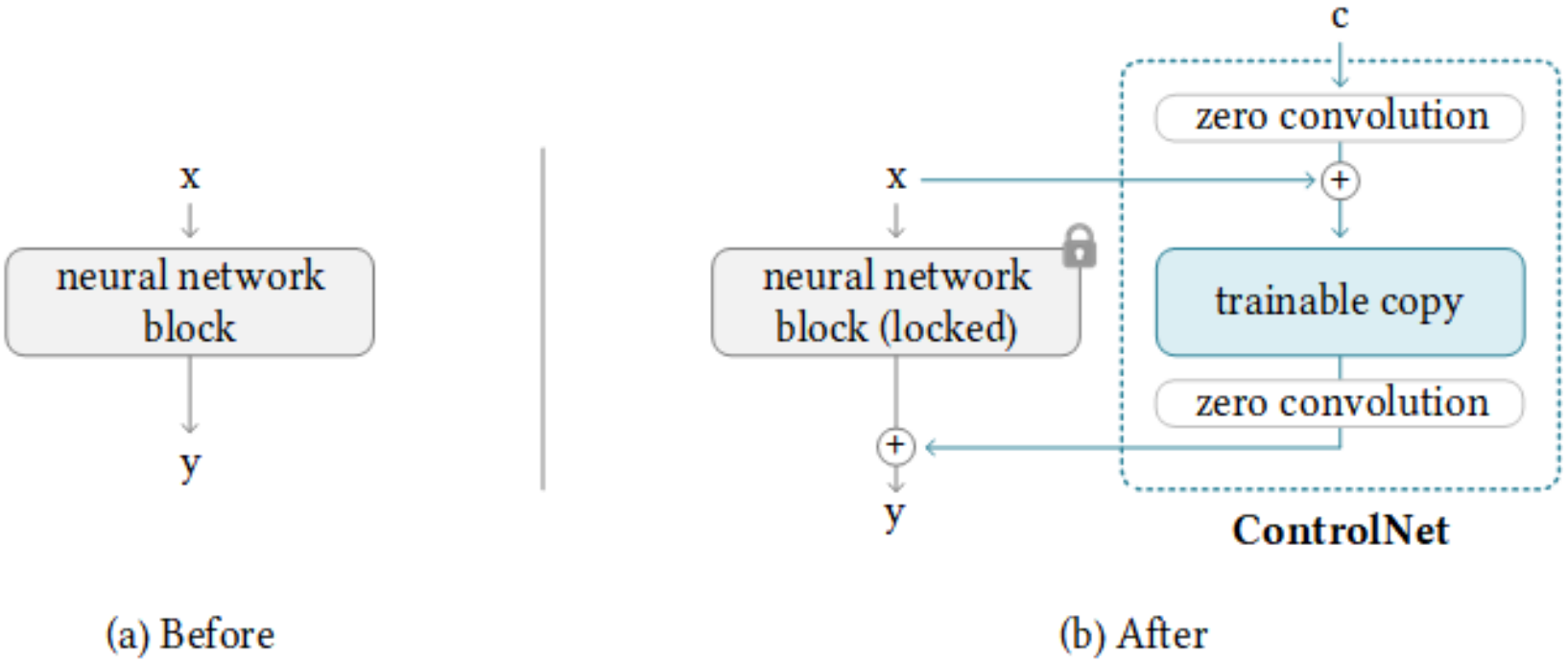

Image source: https://github.com/lllyasviel/ControlNet

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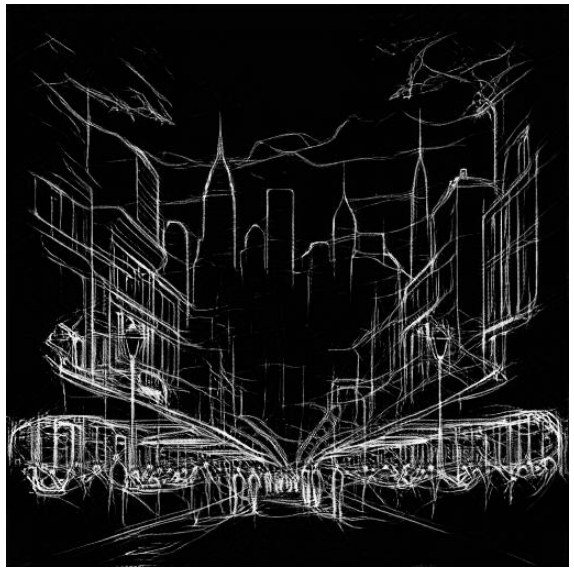
Contour ControlNet Models

- Contour ControlNet models restrict the image content based on the element contour.
- Contour ControlNet models include Canny, MLSD, Lineart, Lineart_anime, SoftEdge, Segmentation, and Shuffle.

Canny

MLSD



Lineart

...

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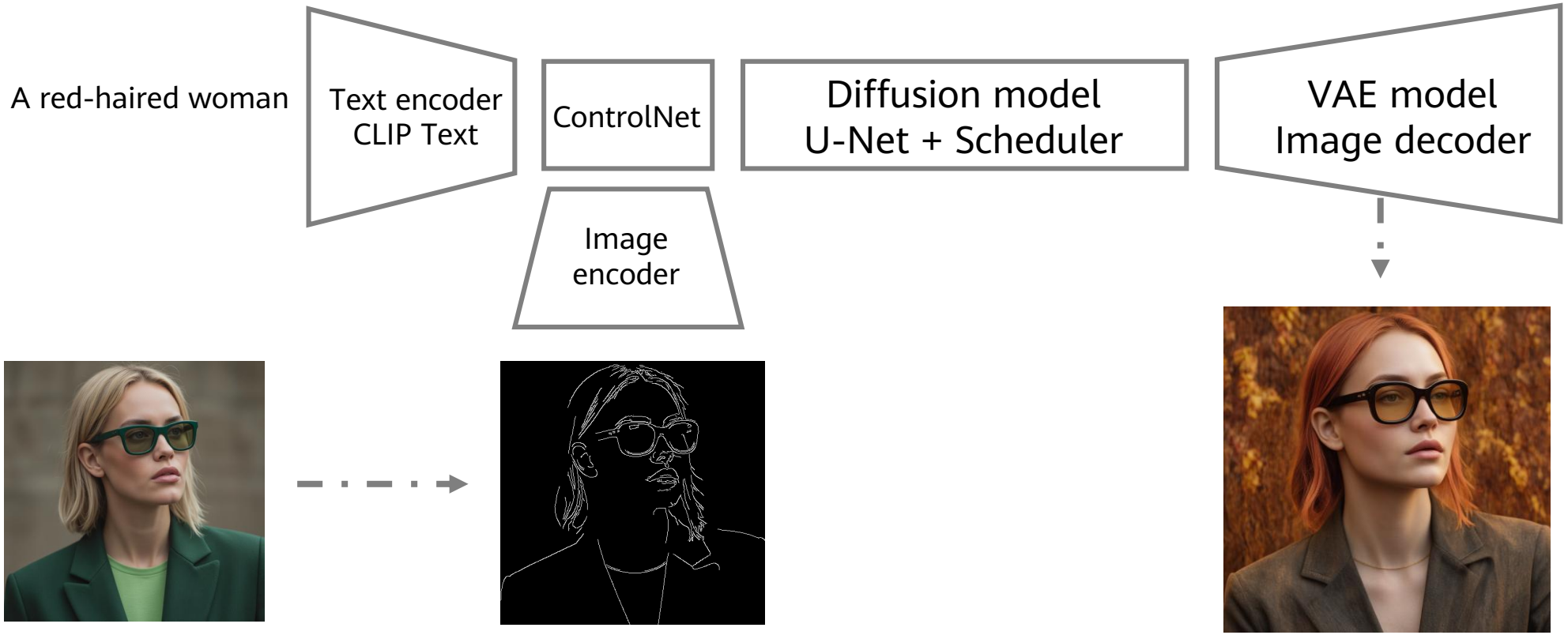
Canny

- The Canny model is derived from the edge detection algorithm in image processing. It can identify and extract edge features from the source image and transfer them to the generated image. It works with the main model to accurately restore the content layout of the source image in the generated image.



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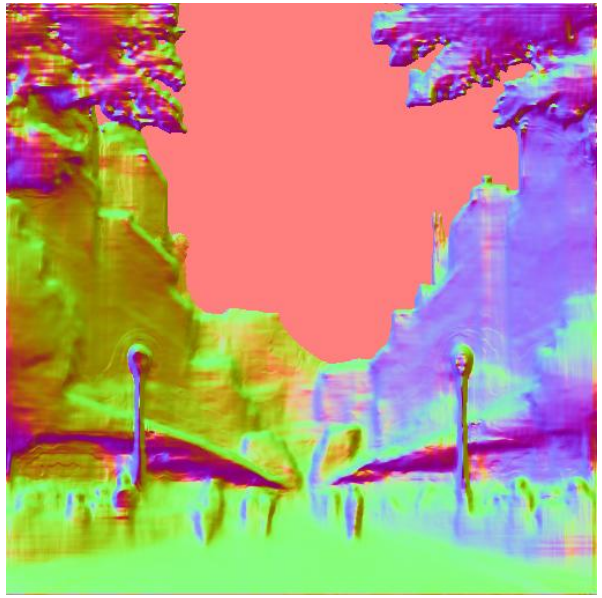
Depth ControlNet Models

- Depth ControlNet models restrict the image content based on the distance between elements and the lens.
- These models reflect the front and back relationships between elements, including Depth and NormalMap.

Depth

Source image



NormalMap

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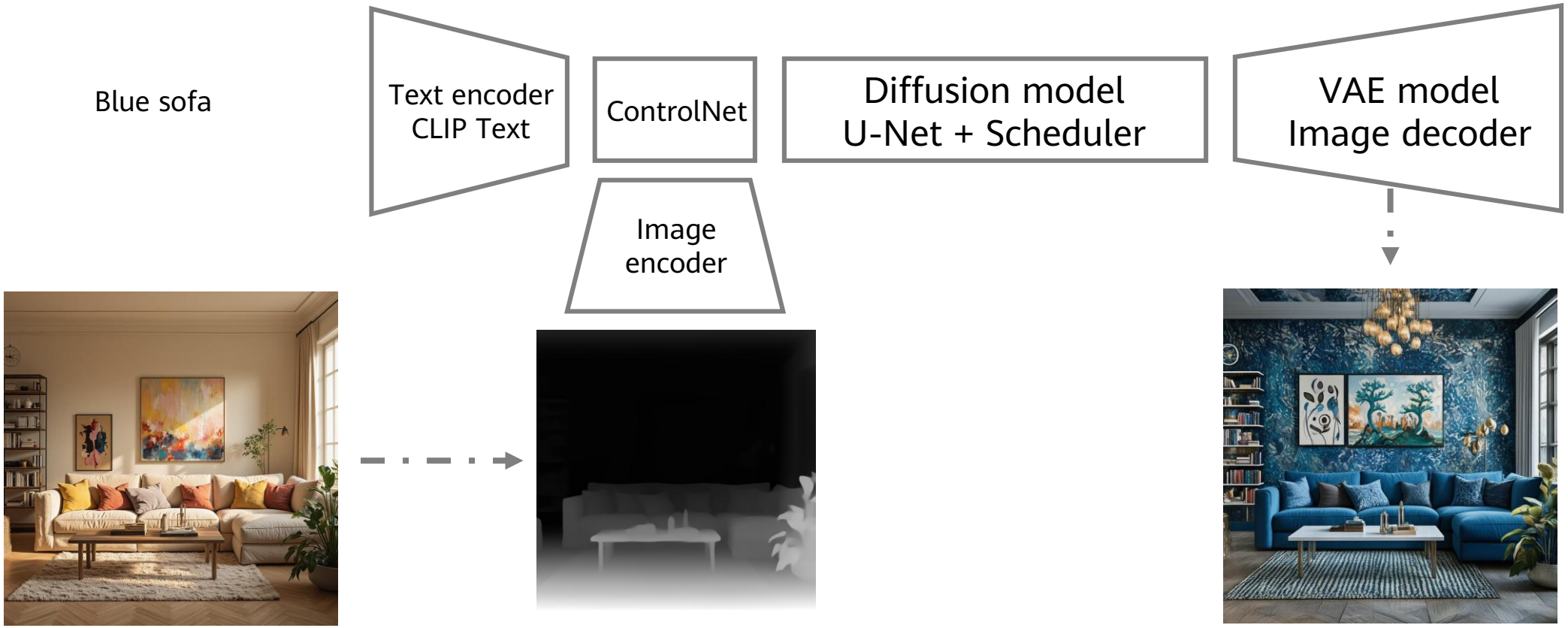
Depth

- A depth map, also known as a distance image, uses the distance (depth) from the image collector to each point in a scene as the pixel value, and reflects the three-dimensional depth relationships between objects in an image. The shorter the distance to the lens, the lighter the color. The longer the distance to the lens, the darker the color.



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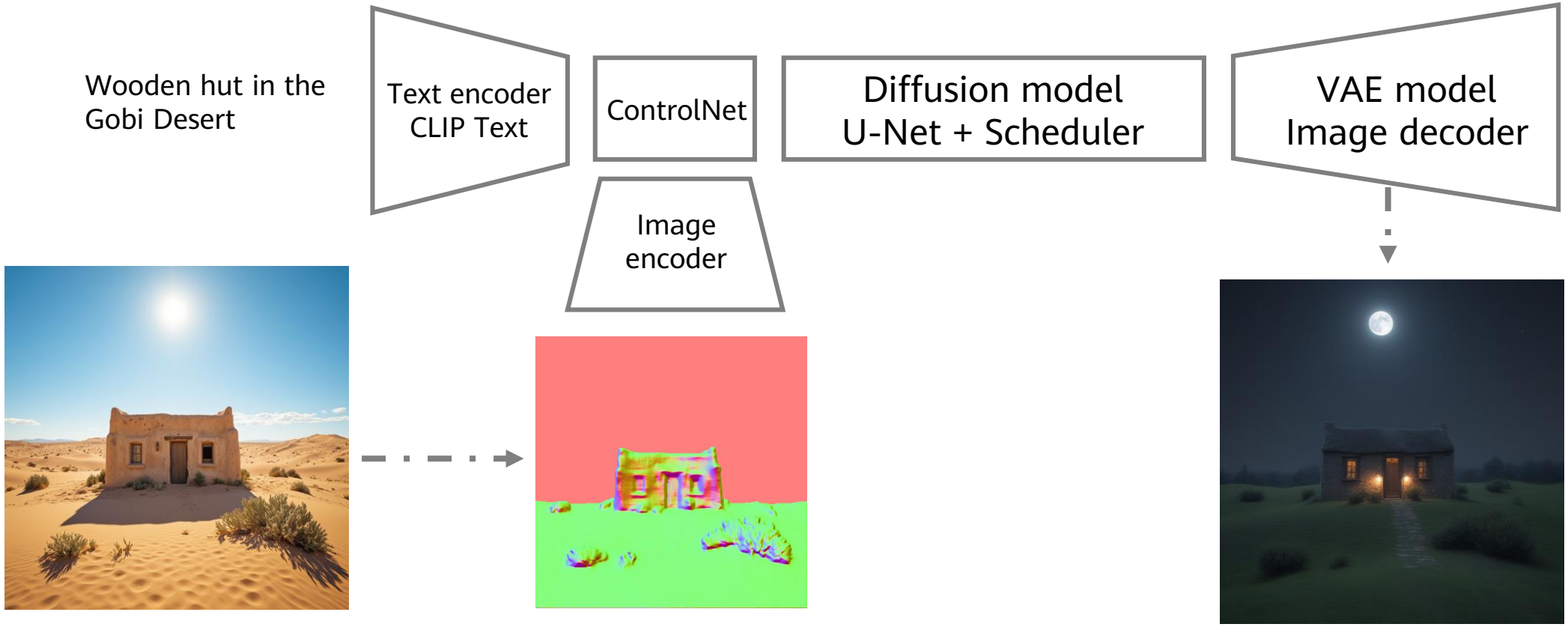
NormalMap

- A normal is a vector perpendicular to a surface. It stores information such as the direction and angle of the surface. A normal is drawn at each point on the bumpy surface of an object and then stored in the RGB color channel, where R, G, and B correspond to the X, Y, and Z axes in a 3D scene, respectively.
- A normal map reflects the light and shadow distribution without changing the structure of an object.



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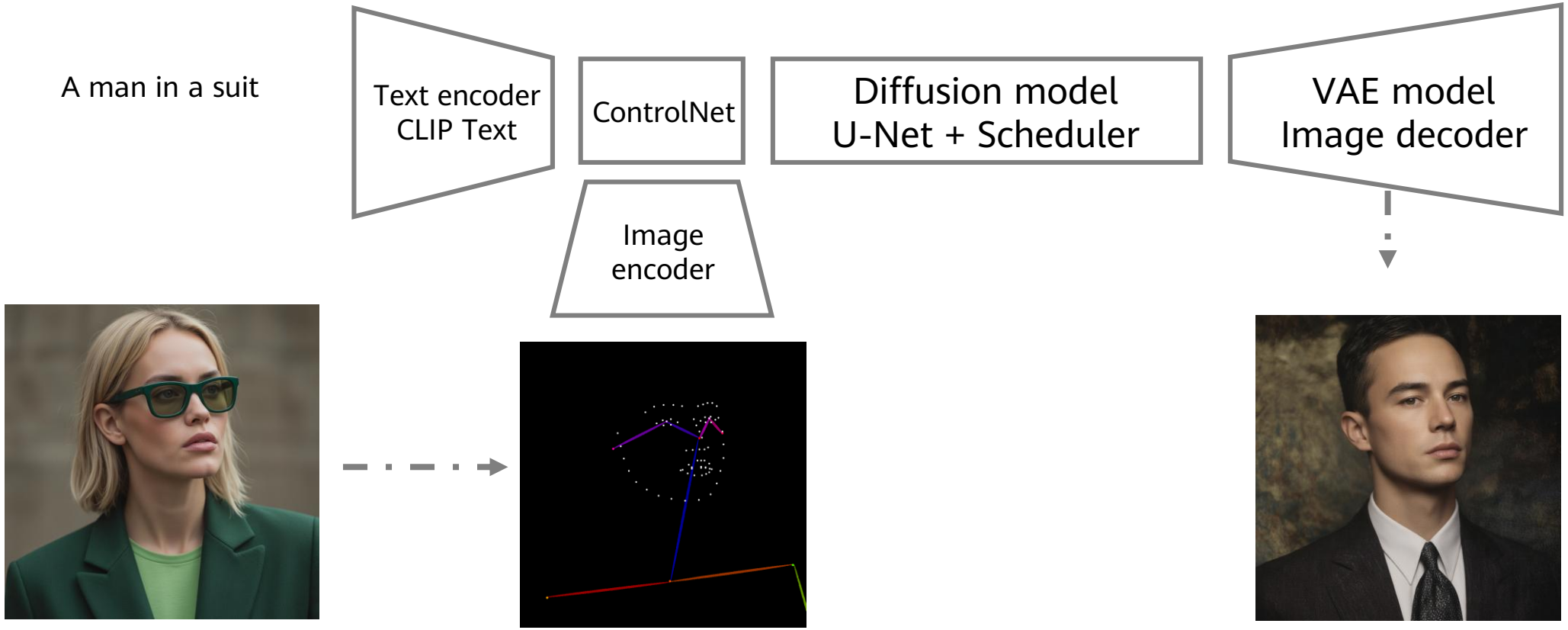
Object ControlNet Models – OpenPose

- OpenPose can detect key parts of a human body, such as the head, shoulders, elbows, and knees, while ignoring details such as clothing, hairstyle, and background. It can capture the position of a human in an image to restore the human's posture and expression.



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Summary

- What is ControlNet
- Common ControlNet models

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Quiz

1. Which of the following is not a contour ControlNet model?

   A. Canny

   B. Lineart_anime

   C. SoftEdge

   D. Depth

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Recommendations

- Huawei Cloud websites
  - Huawei Cloud official website: https://www.huaweicloud.com/intl/en-us/
  - Huawei Cloud Developer Institute: https://edu.huaweicloud.com/intl/en-us/

Huawei Cloud Developer Institute

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